

Which table shows the combination of expected hormones present in a female two weeks after ovulation if fertilization has occurred?

✔ ☒ A.

High level	Low level
Estrogen Progesterone	FSH LH

[49%]

☐ B.

High level	Low level
FSH LH	Estrogen Progesterone

[10%]

☐ C.

High level	Low level
Progesterone LH	Estrogen FSH

[25%]

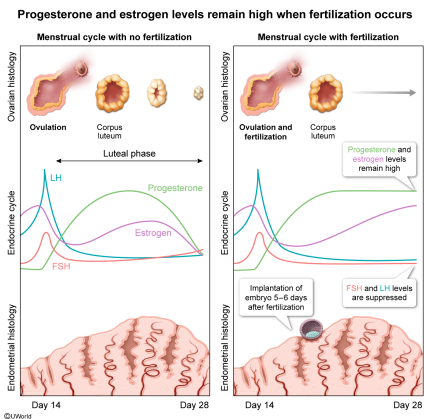
☐ D.

High level	Low level
Progesterone FSH	Estrogen LH

[14%]

Correct
49% answered correctly

Explanation:



Each month, a fertile female undergoes a series of changes that make sexual reproduction possible. The female **menstrual cycle** is controlled by the **endocrine system** and can be divided into three phases:

- **Follicular phase (Days 1–13):** **Follicle stimulating hormone** (FSH) and **luteinizing hormone** (LH) are secreted by the **anterior pituitary gland**. FSH peaks during the first week of the cycle and stimulates the development of several **ovarian follicles**. Subsequently, LH begins to surge, and one follicle becomes dominant. **Estrogen**, produced by the ovaries, also increases during this time. Simultaneously, endometrial changes include menses and the initial stages of endometrial proliferation.
- **Ovulation (Day 14):** Soon after the LH surge, the ovarian follicle matures, and an **oocyte** is released.
- **Luteal phase (Days 15–28):** FSH and LH cause the remnants of the follicle to be converted into the **corpus luteum**, which produces **progesterone**. Progesterone stimulates the thickening of the **endometrium** to make it receptive to embryo **implantation** and suppresses FSH and LH. If fertilization does *not* occur, the corpus luteum degenerates, and progesterone and estrogen levels decline sharply, leading to another **menses** at the beginning of the next cycle.

The question asks for the correct combination of hormones two weeks after ovulation in a female's menstrual cycle if fertilization has occurred. After successful fertilization, hormone levels during the luteal phase are altered. **In a pregnant female**, the corpus luteum does not degenerate, and progesterone and estrogen levels remain high. Because the corpus luteum is still present, the FSH and LH levels remain suppressed. Therefore, two weeks after ovulation, **estrogen and progesterone** levels will be **high** and **FSH and LH** levels will be **low**.

(Choices B, C, and D) Due to the continued presence of the corpus luteum following fertilization, estrogen and progesterone levels would be high (*not* low) two weeks after ovulation. LH and FSH levels would remain *suppressed* because a new menstrual cycle is not initiated.

Educational objective:

During the follicular phase of the menstrual cycle, follicle stimulating hormone (FSH) and luteinizing hormone (LH) influence the maturation of ovarian follicles, resulting in the release of a mature oocyte. During the luteal phase, progesterone and estrogen prepare the endometrium for implantation. Compared to a typical menstrual cycle, the luteal phase in early pregnancy is characterized by high progesterone and estrogen levels, and low FSH and LH levels.

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